Exp. No.: 07
DATE: 10/03/26
RUNO.: 27

ANALYSIS OF MALFUNCTIONING RECTIFIER OR AMPLIFIER CIRCUIT

AIM:

To analyze a malfunctioning rectifier circuit and to identify faults and troubleshoot to restore proper operation.

APPARATUS REQUIRED:

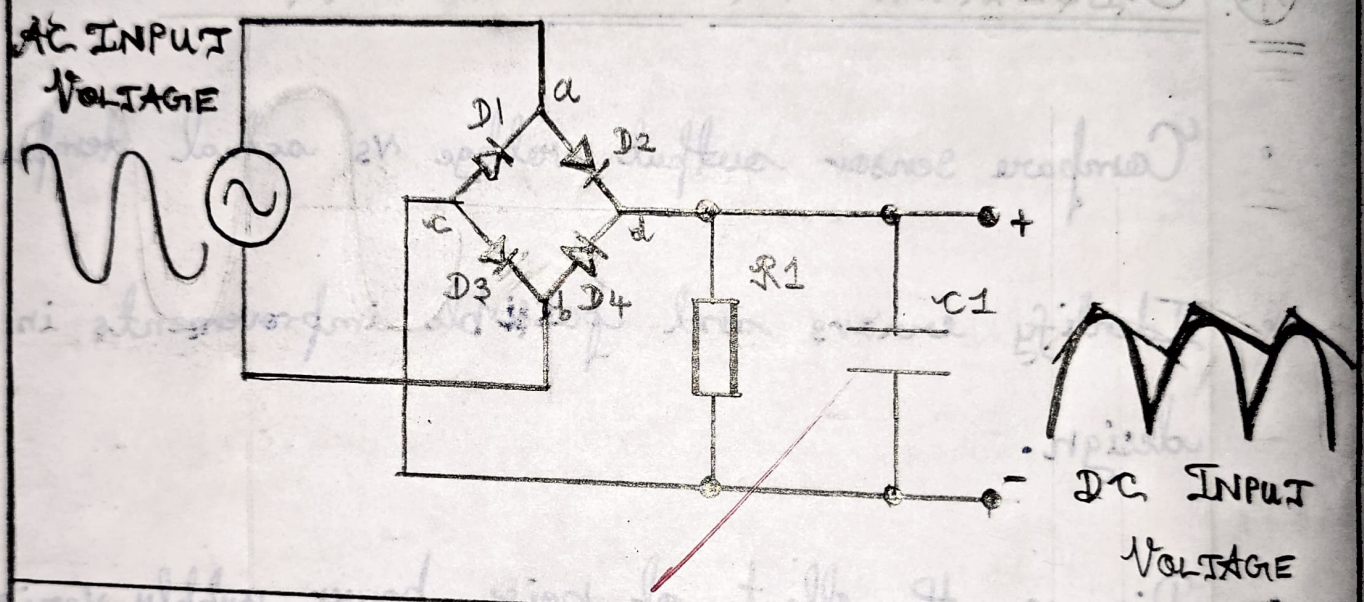
- Faulty rectifier circuit (Bridge or Half-Wave).
- Digital Multimeter (DMM).
- Oscilloscope.
- Function Generator (if needed).
- DC Power Supply.
- Load Resistor.
- Soldering iron and solder.
- Spare diodes, resistors, capacitors.

PROCEDURE:

Step 1: Visual Inspection:-

1. Check for burnt or broken components.
2. Inspect for loose connections or solder joints.

CIRCUIT DIAGRAM:



Step 2: Voltage and Current Testing:-

1. Power off the circuit and measure the continuity of the diodes using a multimeter in the "diode" mode.
   - A good diode should show low resistance in one direction and high resistance in the other.
   - A shorted diode will show low resistance in both directions.
   - An open diode will show no conduction in either direction.
2. Power on the circuit and measure input and output voltages.
   - Compare with expected values for proper rectifier function.

Step 3: Waveform Analysis Using an Oscilloscope:-

1. Connect the oscilloscope - probes across the rectifier output.
2. Observe the waveform shape:-
   - A working rectifier should show a rectified waveform.
   - An open diode may cause missing pulses.
   - A shorted diode may result in A.C. leakage.

Step 4: Checking the Filtering Circuit:-

1. If excessive ripple is observed, test the capacitor using a multimeter in capacitance mode.
2. Replace faulty capacitors if needed.

Step 5: Troubleshooting and Correction:

1. Based on the findings, replace defective diodes or capacitors.
2. Re-solder loose connections and re-test.

⊛. OBSERVATIONS AND RESULTS:

- Note the measured voltages and compare them with theoretical values.
- Record oscilloscope waveforms and analyze irregularities.
- Identify the root cause of the malfunction.

⊛. RESULT:

Thus, the analysis of a malfunctioning rectifier or amplifier circuit was done and the faults were identified. The circuit was troubleshooted to restore proper operation.